Workbook Template

Project Name: **Task management system and reminder**

Problem Statement:

People obtained have many personal, academic, or professional task to complete. Remembering every task and its deadline can be difficult. Manual notes and reminders may not provides proper task tracking.

Why does this problem exist?

People have multiple tasks at the same time.Important deadlines may be forgotten.Manual task lists are difficult to update.There may be no proper system to prioritize tasks. Users may forget upcoming appointments or activities.It is difficult to track completed and pending tasks.Lack of reminders can result in incomplete or delayed work.

Who are the stakeholders?

Internal users:

1. System
2. Administrators
3. Project Managers
4. Registered Users

External users:

1. Students
2. Teachers
3. Employees/Professionals
4. General users

Expected system users:

1. Students
2. Teachers
3. Office employees
4. Professionals
5. Smallteams
6. General users

Objectives of the project:

1. To create and manage tasks easily.
2. To set deadlines for tasks.
3. To provide reminders for important tasks.
4. To categorize tasks according to priority.
5. To track pending and completed tasks.
6. To provide a simple and user-friendly interface.
7. To improve productivity and time management.

Functional requirements:-

1. Create an account and log in.
2. Add new tasks.
3. Enter task title and description.
4. Set task date and time.
5. Set priority such as High, Medium, or Low.
6. Set reminders.
7. Update task status.
8. Delete unwanted tasks.
9. View completed and pending tasks.

Nonfunctional requirements:

1. Usability: The system should be easy to use.
2. Performance: The application should respond quickly.
3. Security: User information should be protected.
4. Reliability: Tasks and reminders should be stored
5.correctly. Maintainability: The system should be easy to update.

Expected outputs:

1. User can register and log in securely.
2. User can add new tasks with title and description.
3. User can set task deadlines with date and time.
4. User can set task priority as High, Medium, or Low.
5. System can set reminders for upcoming tasks.
6. User can edit or delete tasks when required.
7. User can mark tasks as completed or pending.
8. Dashboard displays total, pending, completed, and overdue tasks.

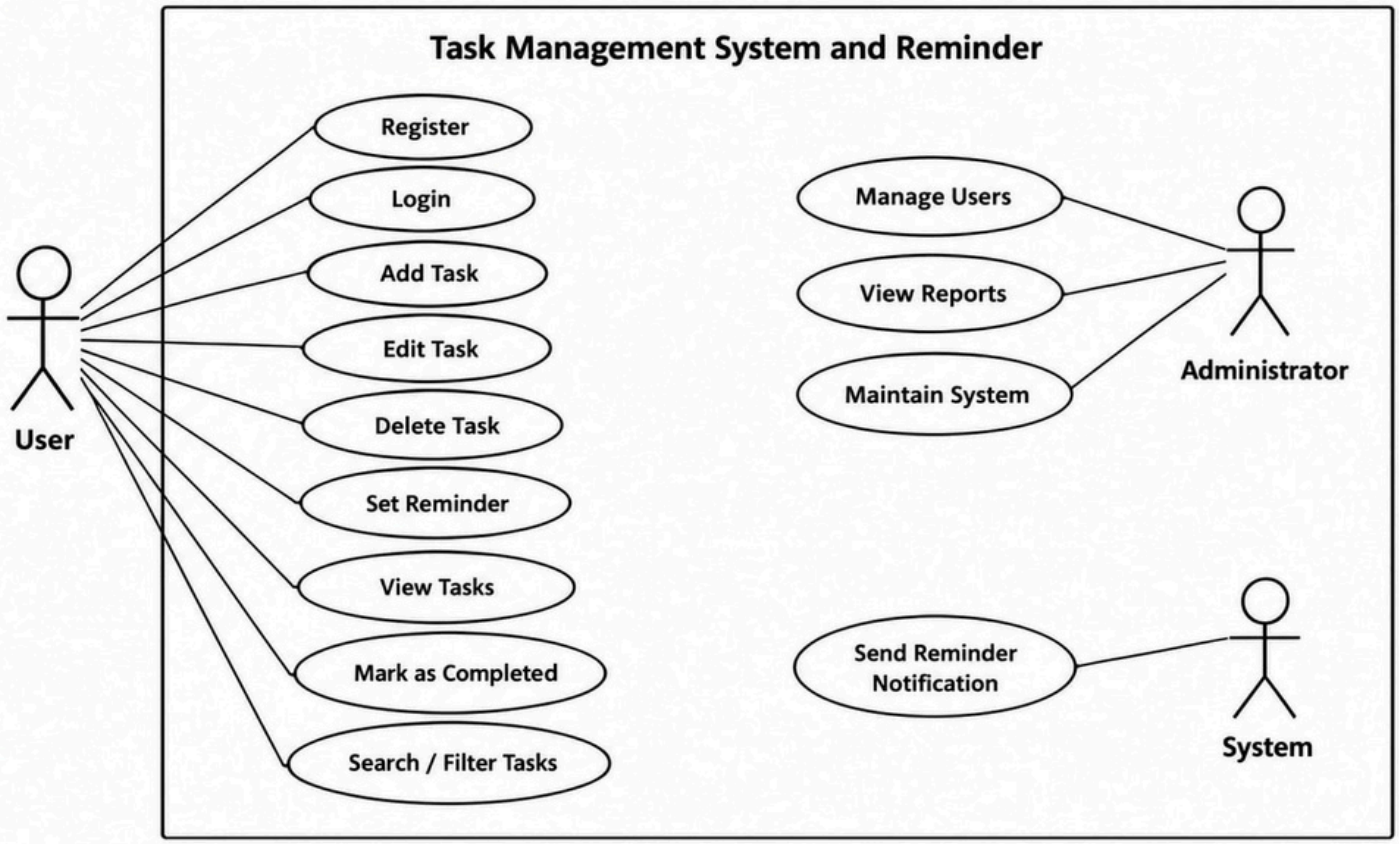
Hands-on activity:

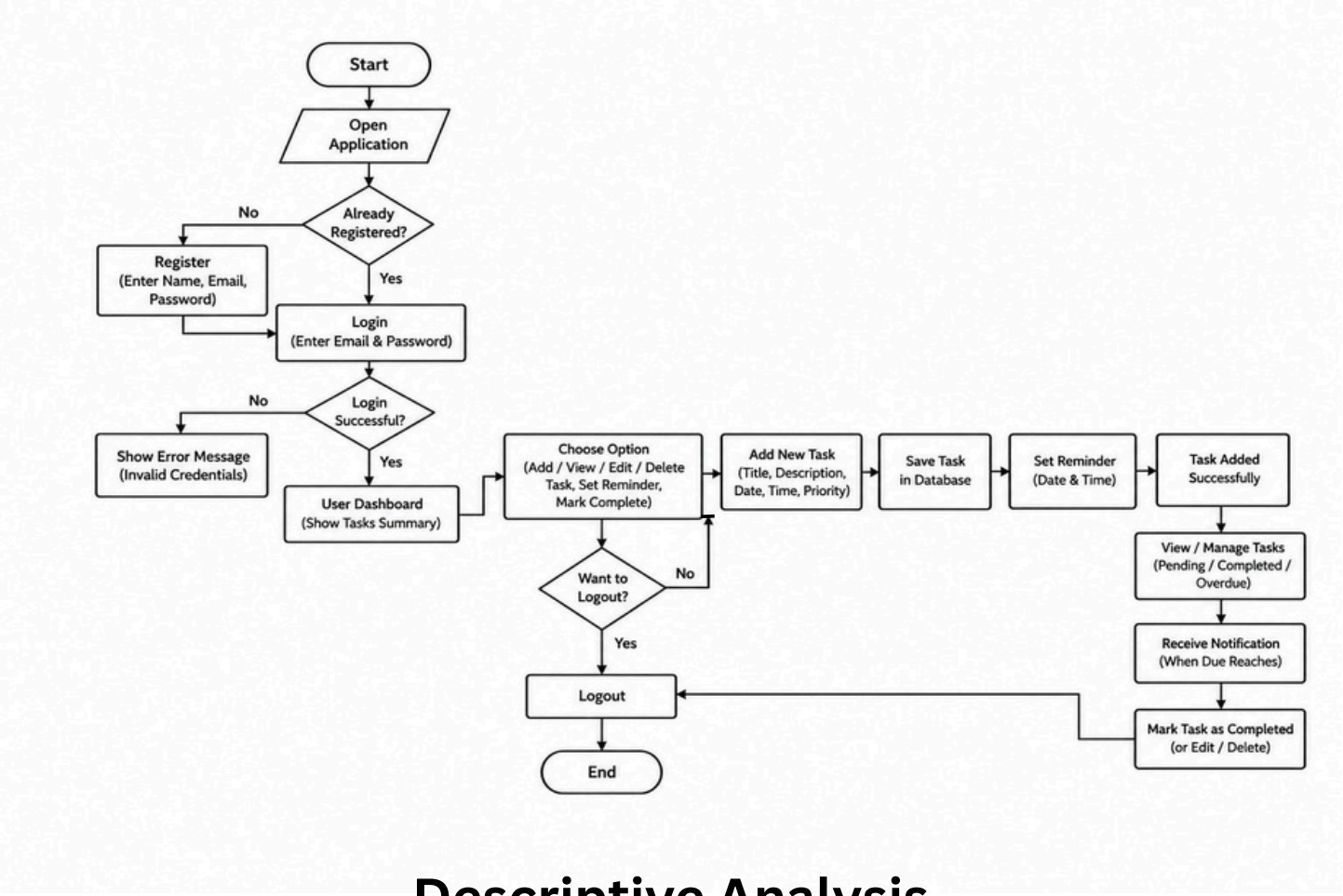
1. Open the Task Management System.
2. Register a new user account.
3. Login using the registered account.
4. Open the Add Task option.
5. Enter the task title, for example, "Complete BCA Assignment."
6. Add a short description of the task.
7. Select the due date.
8. Select the task priority as High, Medium, or Low.
9. Set a reminder date and time.
10. Save the task.
11. Check the task on the dashboard.
12. After completing the work, change the task status to Completed.
13. Verify that the task appears in the completed task section.

MIS components:

The Task Management System and Reminder is an information system designed to help users organize, manage, and track their daily tasks efficiently. The system allows users to create tasks, set deadlines, update task details, mark tasks as completed, and receive reminders for important activities. MIS helps collect and store task-related information and provides useful information to users for better planning and decision-making. The system reduces the chances of forgetting important tasks and improves time management and productivity. It provides accurate and timely information about pending, completed, and upcoming tasks in a simple and organized manner.

1. Use Case Diagram





Descriptive Analysis

Descriptive analysis explains what has happened in the Task Management System. It analyzes completed, pending, overdue, and upcoming tasks. For example, the system can generate reports showing the number of tasks completed by a user in a week and the number of overdue tasks.

Diagnostic Analysis

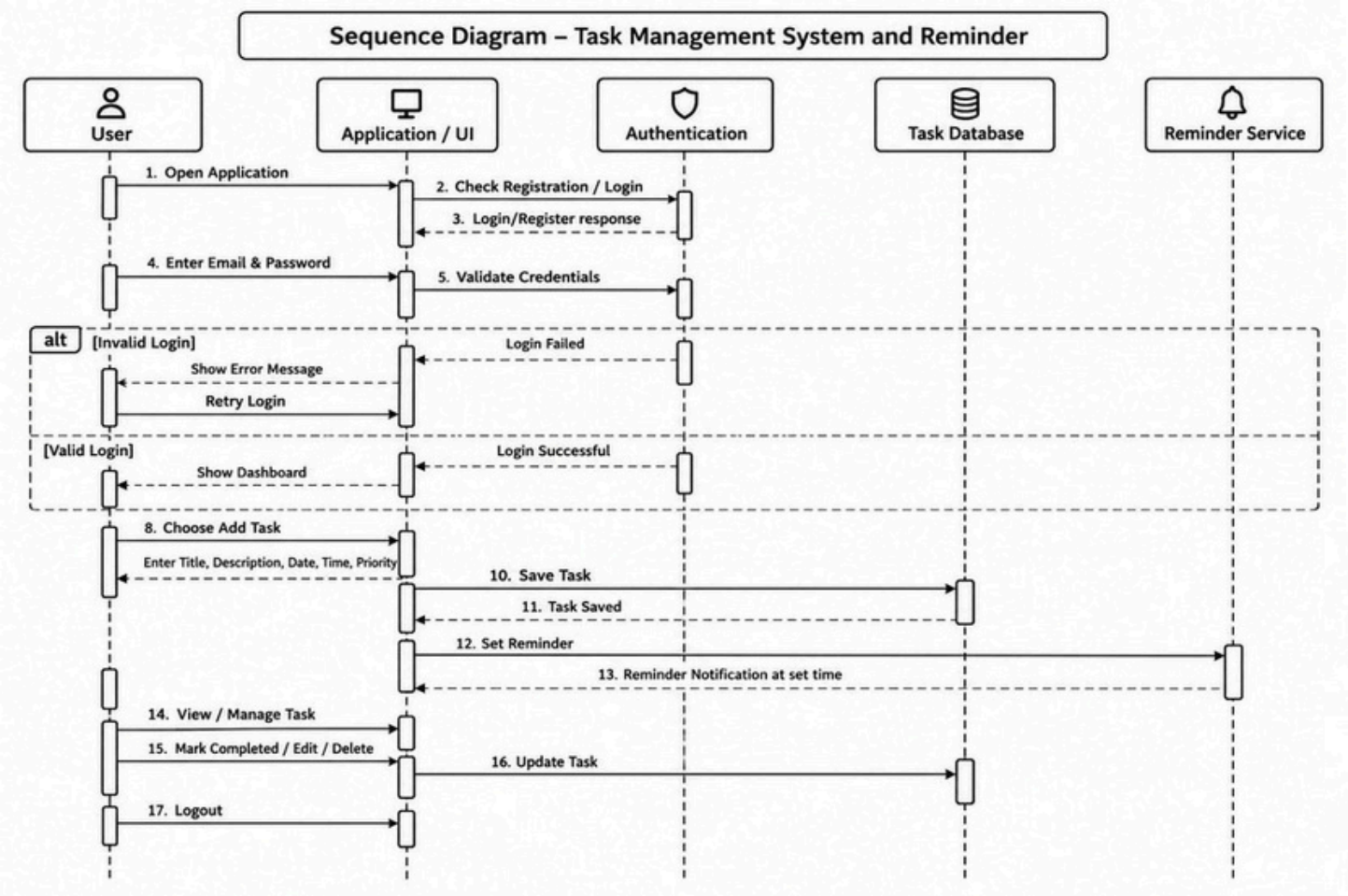
Diagnostic analysis explains why something happened. In this system, it can identify reasons for missed or delayed tasks, such as too many assigned tasks, missed reminders, or incorrect deadlines. This helps users understand the causes of poor task completion.

predictive Analysis

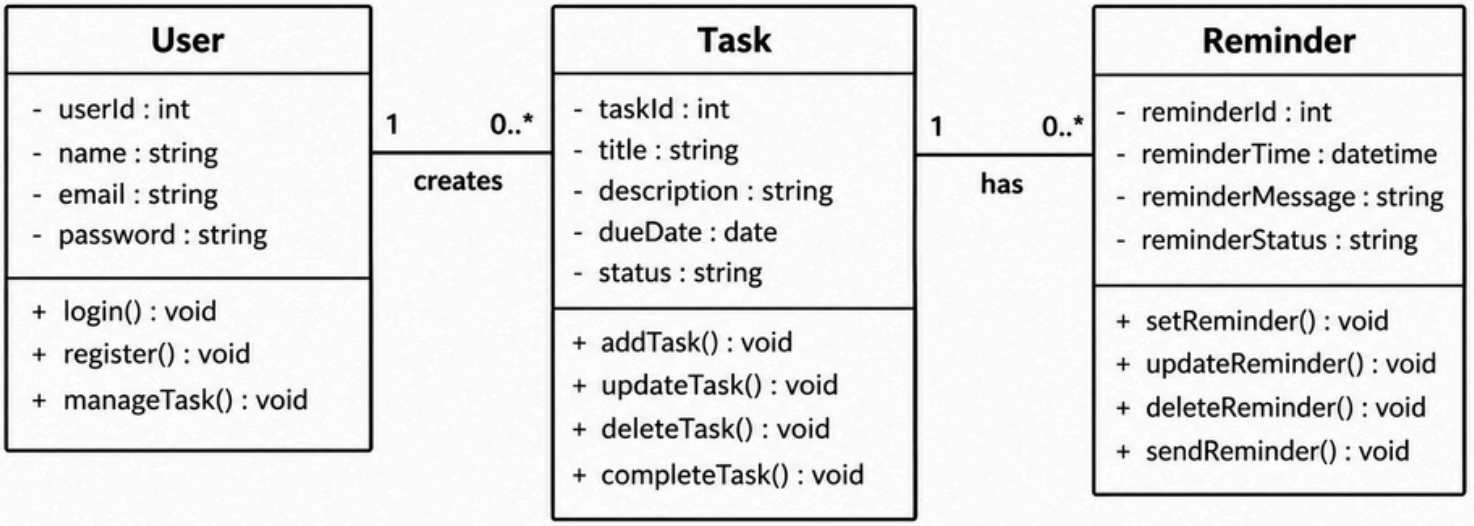
Predictive analysis determines what is likely to happen in the future using previous task data. The system can predict whether a task may become overdue based on the user's past completion patterns, workload, and deadlines. It can also predict periods when the user may have a high workload.

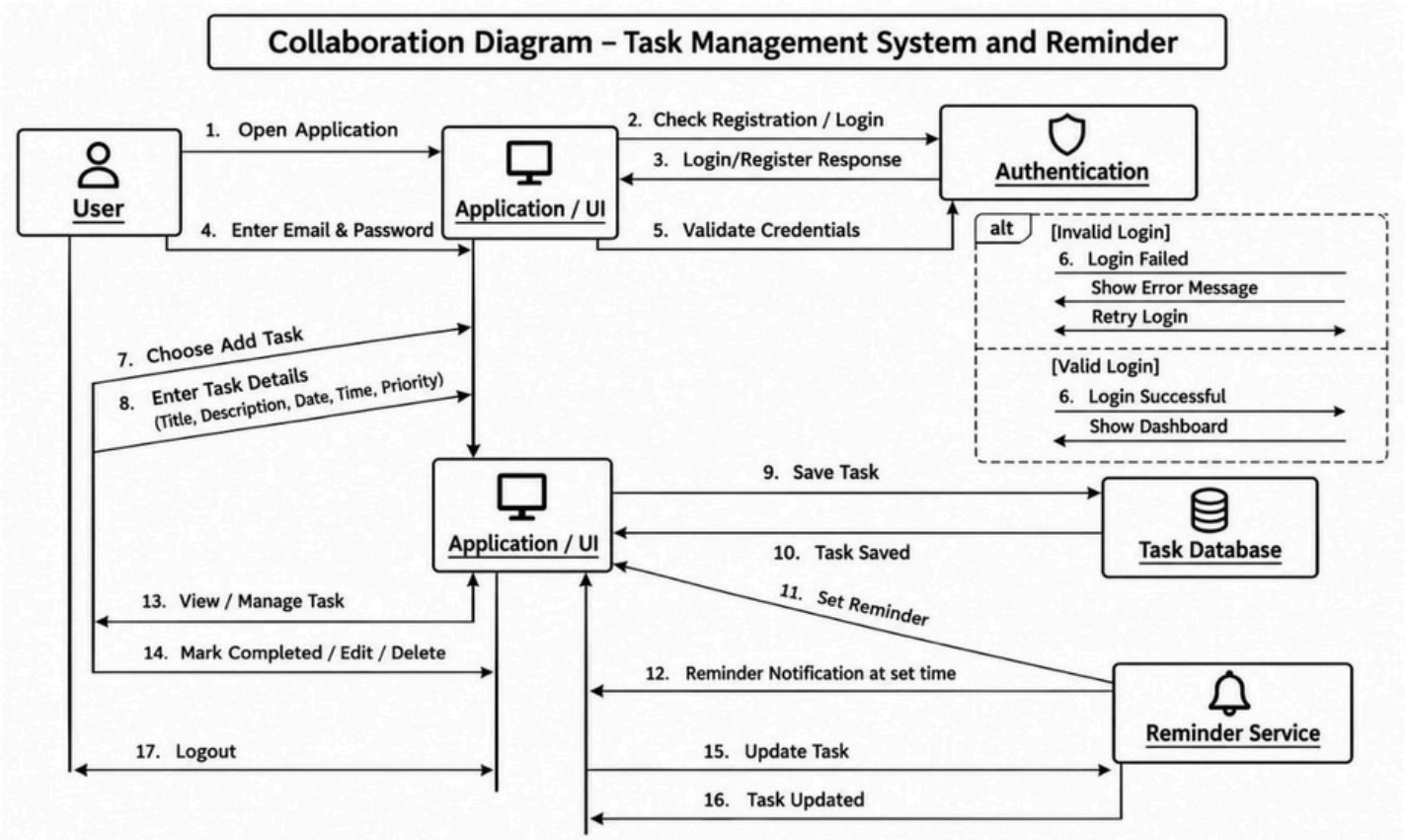
Prescriptive Analysis

Prescriptive analysis suggests what action should be taken. The system can recommend prioritizing urgent tasks, setting earlier reminders, rescheduling low-priority tasks, or dividing large tasks into smaller activities. This helps users manage their time effectively and improve productivity.



Task Management System and Reminder
Class Diagram




Collaboration Diagram – Task Management System and Reminder
User
Application / UI
Authentication
Application / UI
Task Database
Reminder Service
1. Open Application
2. Check Registration / Login
3. Login/Register Response
4. Enter Email & Password
5. Validate Credentials
alt
[Invalid Login]
6. Login Failed
Show Error Message
Retry Login
[Valid Login]
6. Login Successful
Show Dashboard
7. Choose Add Task
8. Enter Task Details
(Title, Description, Date, Time, Priority)
9. Save Task
10. Task Saved
11. Set Reminder
12. Reminder Notification at set time
13. View / Manage Task
14. Mark Completed / Edit / Delete
15. Update Task
16. Task Updated
17. Logout